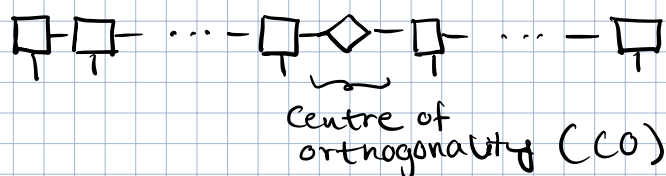


Mixed canonical forms

- * Left canonical to the left of the centre of orthogonality
- * Right — n — right — n

Three types:

$$(1) A^{s_1} A^{s_2} \dots A^{s_n} \sum_n B^{s_{n+1}} \dots B^{s_L}$$



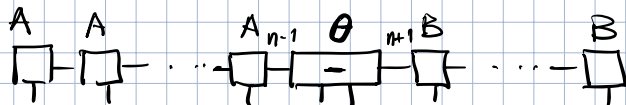
$$|\psi_{(1)}\rangle = \sum_{a_n} \sigma_{a_n} |L_{a_n}^{[1:n]}\rangle \otimes |R_{a_n}^{[n+1:L]}\rangle$$

- * Schmidt decomposition across bond n .
- * Can read off entanglement spectrum/entropy
- * Truncation is transparent.
- * Disadvantage: no physical index on CO

$$(2) : |\psi_{(2)}\rangle = \sum_{a_{n-1} a_n s_n} \underbrace{A_{a_{n-1} a_n}^{s_n}}_{\text{CO}} \sum_n |L_{a_{n-1}}^{[1:n-1]}\rangle |s_n\rangle |R_{a_n}^{[n+1:L]}\rangle$$

- * CO has one physical index
 $\Rightarrow$ Local operator $\hat{O}_n$ can be easily evaluated.

$$(3): |\psi_{(3)}\rangle = \sum_{a_{n-1} a_{n+1} s_n s_{n+1}} \theta_{a_{n-1} a_{n+1}}^{s_n s_{n+1}} \underbrace{|\underbrace{L_{a_{n-1}}^{[1:n-1]}}_{A^{s_1} A^{s_2} \dots A^{s_{n-1}}} \rangle}_{A^{s_1} A^{s_2} \dots A^{s_{n-1}} | s_1 s_2 \dots s_{n-1} \rangle} |s_n\rangle |s_{n+1}\rangle |R_{a_{n+1}}^{[n+2:L]}\rangle$$



$$\dim \theta = (\chi_d, \chi_d)$$

↑ for truncated MPS

$\Rightarrow$ we can increase bond dimension on bond n up to $\chi \cdot d$

$$\langle L_{a_i} | L_{a_j} \rangle = \delta_{a_i a_j}$$

$$\langle R_{a_i} | R_{a_j} \rangle = \delta_{a_i a_j}$$

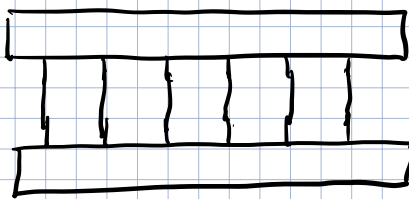
$$U^\dagger U = \mathbb{1}$$

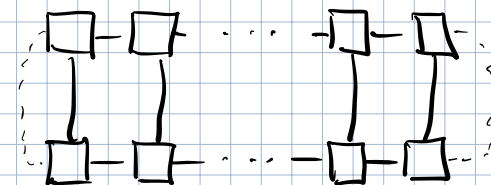
$$V^\dagger V = \mathbb{1}$$

Contractions: norms, overlaps

Canonical forms allow for easy evaluation of norms and expectation values of local operators.

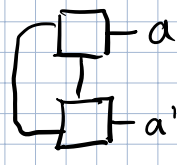
Reason: Need to contract only a few local tensors.

$$\langle \psi | \psi \rangle =$$


$$=$$


Show in exercises:

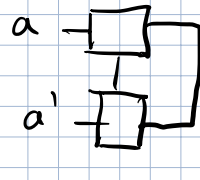
Left-canonical:
(contracted over
left indices)



$$= \begin{bmatrix} a \\ a' \end{bmatrix} = \delta_{a'a} \quad (1)$$

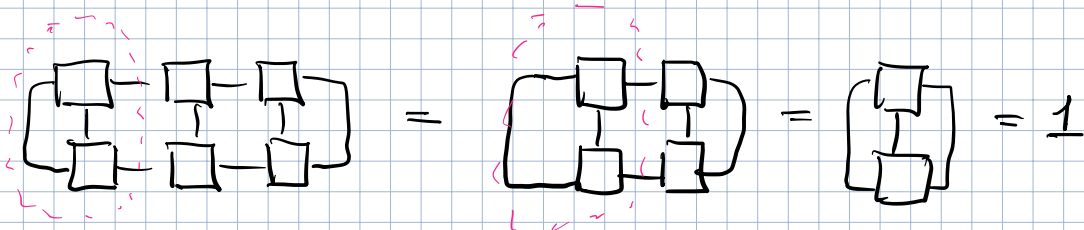
Right can.

?



$$= \begin{bmatrix} a \\ a' \end{bmatrix} = \delta_{a'a}$$

If $|\psi\rangle$ in left-can. form:



$\Rightarrow$ Normalized!

$$\begin{aligned}
 \langle \psi_{(1)} | \psi_{(1)} \rangle &= \text{Diagram with two rows of squares connected by vertical lines, with diamonds in the middle of each column.} \\
 &= \text{Diagram with two diamonds connected by two vertical lines.} = \text{Tr}(\Sigma_n^+ \Sigma_n) = \sum_{a_n} \sigma_{a_n}^2
 \end{aligned}$$

$$\langle \psi_{(3)} | \psi_{(3)} \rangle = \text{Diagram with two rows of squares, each containing a circle with a theta symbol, connected by vertical lines.} = \sum_{s_n s_{n+1}} \text{Tr}(\theta^{s_n s_{n+1}} + \theta^{s_{n+1} s_n})$$

Norm of mixed canonical forms can be read off the CO.

For a local operator on site n $\hat{O}^{[n]}$

$\hat{O}^{[n]}$ has two physical indices $\sim |s_n\rangle \langle s'_n|$

$$\langle \psi | \hat{O}^{[n]} | \psi \rangle = \text{Diagram with a vertical stack of three squares. The top square is labeled A^n. The middle square contains a circle with an O. The bottom square is connected to the top one by a vertical line. The entire stack is enclosed in a large bracket on the right.}$$

Matrix Product Operators (MPO)

Express the Hamiltonian as a product of local operators.

$$H = \sum_{\{s, s'\}} W_1^{s_1 s'_1} W_2^{s_2 s'_2} \dots W_L^{s_L s'_L} |s_1 \dots s_L\rangle \langle s'_1 \dots s'_L|$$

(one s and one s'
for each site $1, \dots, L$)

$W_i^{s_i s'_i}$ has bond dimensions $D_{i-1} \times D_i$

So W_i is a $D_{i-1} \times D_i$ matrix, where each element is a $d \times d$ operator acting on site i .
(W_i is a matrix of local operators)

Total dim is $d D_{i-1} \times d D_i$

$$H|4\rangle = \left(\begin{array}{c} \square - \square - \square - \square \\ | \quad | \quad | \quad | \\ \square - \square - \square - \square \end{array} \right) |4\rangle$$

$\hat{H}$

Giving a new MPS of bond dim. χD .

d : dimension of local Hilbert space $d = \uparrow, \downarrow$
 $d = 0, 1$

r_i : Schmidt-rank across bond i (bond dimension)

χ : truncated bond dimension

$D_i: "r_i"$, but for operators

Vertical lines: sum over $s_i = 1, \dots, d$
(physical index)

Horizontal lines: sum over MPS / MPO indices.

$$a_i = 1, \dots, x$$

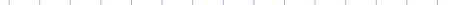
$$a_i = 1, \dots, D_i$$

$$| \psi \rangle = \sum_{s_1 \dots s_L} C_{s_1 \dots s_L} | s_1 \dots s_L \rangle$$

$$\dim C = d^2$$

Example: Heisenberg model

Nearest-neighbour coupling along a chain.

$$H = J \sum_i S_i \cdot S_{i+1}$$


6 sites

$$= J \sum_i (S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + S_i^z S_{i+1}^z)$$

2 sites: 

$$W_1 = \begin{pmatrix} \cancel{1} & s_1^x & s_1^y & s_1^z & \cancel{0} \end{pmatrix} \quad \vec{s}_1^T$$

$$W_1 W_2 = J \vec{S}_1 \cdot \vec{S}_2 = H$$

For a chain of length L :

$$\text{Keep } W_1 = (1 \quad \underbrace{S_1^x \quad S_1^y \quad S_1^z}_{\text{Local operator, } d \times d \text{ matrix}} \quad 0) , \quad W_L =$$

$$\begin{pmatrix} 0 \\ J S_L^x \\ J S_L^y \\ J S_L^z \\ 1 \end{pmatrix}$$

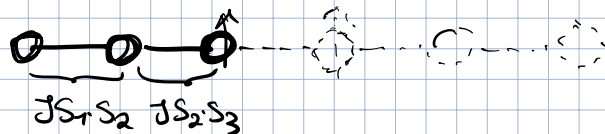
In bulk:

$$W_i = \begin{pmatrix} 1 & S_i^x & S_i^y & S_i^z & 0 \\ 0 & 0 & 0 & 0 & J S_i^x \\ 0 & 0 & 0 & 0 & J S_i^y \\ 0 & 0 & 0 & 0 & J S_i^z \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$D_i = 5$$

$$W_1 W_2 = (1 \quad S_1^x \quad S_1^y \quad S_1^z \quad J S_1 \cdot S_2)$$

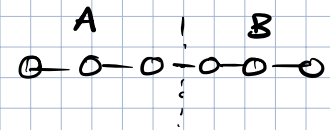
$$W_1 W_2 W_3 = (1 \quad S_3^x \quad S_3^y \quad S_3^z \quad J(S_1 \cdot S_2 + S_2 \cdot S_3))$$



$$\underline{W_1 W_2 \dots W_L = J \sum_i S_i \cdot S_{i+1} = H}$$

Why is $D_i = 5$?

Cut H in two at bond i :



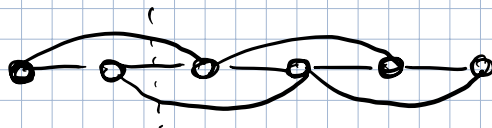
$$H = H_A \otimes \mathbb{1} + \mathbb{1} \otimes H_B + \underbrace{J S_i^x S_{i+1}^x + J S_i^y S_{i+1}^y + J S_i^z S_{i+1}^z}_{5 \text{ terms!}}$$

$$W_A = W_1 W_2 \dots W_i = (\mathbb{1} \ S_i^x \ S_i^y \ S_i^z \ H_A)$$

$$W_B = W_{i+1} \dots W_L = \begin{pmatrix} H_B \\ J S_L^x \\ J S_L^y \\ J S_L^z \\ \mathbb{1} \end{pmatrix}$$

(First - and)

Second - neighbour:



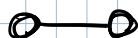
$$H = J_1 \sum_i S_i \cdot S_{i+1} + J_2 \sum_i S_i \cdot S_{i+2}$$

$$W_1 = (\mathbb{1} \ S_1^x \ S_1^y \ S_1^z \ 0 \ 0 \ 0 \ 0) \quad -h \cdot S_i^z$$

↑ This could be non-zero for on-site interaction e.g. magnetic field

$$W_i = \begin{pmatrix} 1 & S_i^x & S_i^y & S_i^z & 0 & 0 & 0 & -h S_i^z \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & J_1 S_i^x \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & J_1 S_i^y \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & J_1 S_i^z \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & J_2 S_i^x \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & J_2 S_i^y \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & J_2 S_i^z \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \quad -h \cdot S_i^z$$

$$W_1 W_2 = (1 \quad S_1^x \quad S_1^y \quad S_1^z \quad S_2^x \quad S_2^y \quad S_2^z \quad J_1 S_1 \cdot S_2)$$



$$W_1 W_2 W_3 = (1 \quad S_1^x \quad S_1^y \quad S_1^z \quad S_2^x \quad S_2^y \quad S_2^z \quad J_1(S_1 \cdot S_2 + S_2 \cdot S_3) + J_2(S_1 \cdot S_3))$$

Fermions

The MPS describes distinguishable sites, the ansatz does not contain / account for fermionic statistics.

Antisymmetry must be put in by hand

We have chosen an occupation basis

$$|n_1 n_2 \dots n_L\rangle$$

$$|1 \ 1 \ 0 \ 0 \ 1\rangle = c_1^\dagger c_2^\dagger c_5^\dagger |0\rangle \quad \{c_i^\dagger, c_j^\dagger\} = 0$$

When an operator, say $c_3^\dagger c_1$, acts on this state, it must be anticommutated through the creation operators to get to site 1 and 3.

$$\begin{aligned} c_3^\dagger c_1 c_1^\dagger c_2^\dagger c_5^\dagger |0\rangle &= c_3^\dagger c_2^\dagger c_5^\dagger |0\rangle \\ &= -c_2^\dagger c_3^\dagger c_5^\dagger |0\rangle \\ &= -|0 \ 1 \ 1 \ 0 \ 1\rangle \end{aligned}$$

Compute overlaps:

$$\langle n_1 n_2 \dots n_L | \dagger | n'_1 n'_2 \dots n'_L \rangle$$

and keep track of minus signs.